

Education

[Islamic University of Technology \(IUT\)](#) [🔗](#)

Bachelor of Science in Computer Science and Engineering (CSE)
CGPA: 3.72/4.0

Jun 2021 – October 2025

Gazipur, Bangladesh

Research

[🔗 Weakly Supervised Semantic Segmentation With Image Labels](#) [🔗](#)

January 2024 – October 2025

I designed a transformer-based approach to enhance object detail recognition in weakly supervised segmentation. Using UniCL with a Swin Transformer, I improved class activation map (CAM) quality and fine-grained feature extraction via windowed attention. My method incorporated advanced CAM refinement, affinity calculations, and encoder intermediate features. This model generated highly detailed masks for most classes, though accuracy dropped for a few categories like person and chair. Achieving a mean IoU of 50%, the approach delivered substantial fine-grained segmentation improvements, despite being below the state-of-the-art result of 74%.

Experience

[🔗 Battery Low Interactive](#) [🔗](#) – Game Development Trainee

Oct 2024

During my industrial training as a Game Developer Trainee, I learned to conceptualize and program games from inception using Unity and C#. I engineered several mini-games to grasp mechanics, scoring algorithms, and physics-driven interactions. For my final project, I developed a 2D centrifugal-force game in which players maneuvered a stone to strike targets, implementing custom game logic, score tracking, and responsive UI components. I also integrated animations, curated assets, and designed scenes. I also explored the fundamentals of AR/VR during that time.

Skills

Programming Languages: Python, C++, JavaScript, TypeScript, Dart, Java, C#, C, SQL

Frameworks & Libraries: PyTorch, YOLOv5, Transformers (Hugging Face), OpenCV, NumPy, Matplotlib, PIL, React, Next.js, Node.js, Flutter, FastAPI, Tailwind CSS

Tools & Technologies: Git, Docker, MongoDB, MySQL, Redis, Jupyter

Projects

[🔗 End-to-End Semantic Segmentation System](#) [🔗](#)

2026 — PyTorch, Hugging Face, FastAPI, Docker, React

- Built and deployed a complete semantic segmentation pipeline from training to real-time inference
- Implemented U-Net and DeepLabV3+ from scratch and integrated SegFormer (Hugging Face) for comparison
- Trained on Carvana and Cityscapes datasets; evaluated using mIoU and pixel accuracy
- Designed modular backend with FastAPI and Docker; exposed REST APIs for inference
- Developed React frontend for image upload and segmentation visualization

[🔗 Sign Tutor AI](#) [🔗](#)

2025 — Python, PyQt6, YOLOv5, OpenCV, PyTorch, NumPy

- AI-powered sign language learning app with real-time YOLOv5; finetuned on custom dataset.
- Gamified learning experience with instant feedback on user performance
- PyQt6 desktop interface with camera-based gesture recognition

[🔗 Blog Writing Agent](#) [🔗](#)

2026 — Python, FastAPI, LangGraph, LangChain, Docker, Pytest

- Built graph-based LLM pipeline (LangGraph: orchestrator-worker-reducer)
- Implemented dual LLM routing (Ollama local, OpenRouter cloud)
- Developed FastAPI backend with modular design and Pydantic validation
- Designed structured generation via task decomposition; containerized with Docker and tested via pytest

[🔗 PlayClassix](#) [🔗](#) ([Live demo](#) [🔗](#))

2026 — Next.js, TypeScript, Tailwind, Zustand, Pusher, Upstash Redis

- Browser-based gaming platform with mini-games (Memory Match, Chess, Wordle)
- Real-time multiplayer via WebSockets with anonymous no-login entry
- Session-persistent user IDs and responsive cross-device design

[🔗 Photo Wizard](#) [🔗](#)

2024 — NumPy, PyQt6

- Desktop image editor with all basic features (crop, resize, blur, sharpen, built-in filters, custom filter creation, etc.)
- All image processing algorithms implemented from scratch using NumPy

Programming Achievements

[IUT Intra First-Year Programming Contest 2021](#) [🔗](#)

2021 — 1st out of 32 teams

[CodeRush 1.0: Programming Competition](#) [🔗](#)

2023 — 7th out of 37 teams